Type 1: Research Project

Explainable Federated Analytics for Multivariate Time-Series Forecasting with Personalized Drift Adaptation

Team Number: 28

S.No	Register Number	Student Name
1	CB.SC.U4CSE23363	Haswitha K
2	CB.SC.U4CSE23529	Hasini Reddy M
3	CB.SC.U4CSE23547	Sheela Akshar Sakhi
4	CB.SC.U4CSE23761	Kousik sarma lakkaraju
5	CB.SC.U4CSE23753	V.Chakravarthy

Guide: **Dr. G R RAMYA ,Dr. VANDHANA S**

Department of CSE

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Problem Statement

To develop a privacy-preserving time-series forecasting system that simultaneously addresses client heterogeneity and concept drift, thereby enabling collaborative and trustworthy analytics in real-world distributed environments

However:

- This data is highly sensitive and private
- Different organizations show different behavioral patterns
- Existing systems provide predictions without explaining why

Because of these limitations:

- Data cannot be shared
- A single global model performs poorly
- Decision-making becomes difficult

Motivation & Real-World Relevance

Key Challenges:

- Privacy laws prevent sharing of real-world data (e.g., patient vitals)
- Centralized AI models are not suitable for distributed environments
- Organizations need accurate, explainable predictions
- Behavior changes over time, requiring adaptive systems

Real-world Applications:

- Hospitals predicting patient risk
- Smart grids forecasting energy demand
- Industries predicting machine failure
- Smart cities managing traffic flow

Literature Survey (1–5)

No.	Author/Year	Method Used	Research Focus	Limitations	Research Gap
1	Churaev et al., Elsevier, 2023	MTCNN face detection; EfficientNet-B0; Apache TVM	On-device facial analysis without raw data sharing	No FL; no personalization; no drift handling	Integrate FL with explainable and personalized analytics
2	Jothimurugesan et al., Springer, 2022	CDA-FedAvg; CUSUM drift detection; memory buffers	Concept drift handling in federated learning	High computation; no time-series focus	Drift-aware explainable time-series analytics
3	Arivazhagan et al., AISTATS, 2020	FedPer with shared base and local heads	Personalized federated learning	No drift handling; no analytics layer	Drift-aware personalized forecasting
4	McMahan et al., PMLR, 2017	FedAvg; local SGD; model averaging	Privacy-preserving decentralized training	No personalization or explainability	Extend with personalization and analytics
5	Liu et al., IEEE Access, 2022	Gradient correction; self-distillation	Stable FL under non-IID data	Not explainable; not time-series focused	Add explainable time-series forecasting

Literature Survey (6–10)

No.	Author/Year	Method Used	Research Focus	Limitations	Research Gap
6	IFIP CNSM, 2025	MVFL; vertical FL; feature separation	Multivariate federated forecasting on edge devices	No explainability; no personalization	Explainable and personalized MVFL
7	Elsevier, 2025	FedXAI; SHAP; LIME; FedAvg	Explainable federated intrusion detection	Classification-only; not time-series	XAI for federated time-series forecasting
8	ETH Zurich, 2025	Federated LSTM; fine-tuning	Financial time-series forecasting	Binary prediction only	Multivariate explainable forecasting
9	MDPI, 2025	FL with LLMs; secure aggregation	Clinical decision support	High complexity; no explainability	Explainable clinical time-series analytics
10	Elsevier/ACM, 2024–25	Personalized FL with local heads	Time-series forecasting under non-IID data	High training complexity	Combine personalization + XAI

Literature Survey (11–15)

No.	Author/Year	Method Used	Research Focus	Limitations	Research Gap
11	arXiv, 2024	FL with LLaMA; PEFT; client clustering	Long-term federated time-series forecasting	High compute cost; edge limitations	Adaptive personalization for heterogeneous clients
12	EDBT, 2025	AutoML FL; meta-features; Bayesian optimization	Automated federated forecasting	Univariate only; meta-feature overhead	Multivariate and personalized FL forecasting
13	SECRYPT, 2025	Microaggregation; k-anonymity; FL	Privacy-enhanced FL forecasting	Information loss; limited scalability	Personalization under strong privacy
14	arXiv, 2024	Fed-TREND; synthetic data; heterogeneity-aware FL	Handling non-IID time-series data	Compute overhead; limited personalization	Explainable personalized forecasting
15	AAAI, 2025	Adaptive look-back horizon; intrinsic space	Optimized federated forecasting horizon	Complex theory; limited deployment	Practical real-time horizon adaptation

Literature Survey (16–20)

No.	Author/Year	Method Used	Research Focus	Limitations	Research Gap
16	IEEE TNNLS, 2022	Federated multi-task learning	Personalized federated forecasting	No explainability	Interpretable federated analytics
17	NeurIPS, 2021	Attention encoder–decoder	Interpretable time-series forecasting	Centralized only	Federated explainable forecasting
18	ACM SIGKDD, 2023	Client clustering; similarity FL	Heterogeneous FL forecasting	No per-client personalization	Personalized explainable FL
19	Elsevier, 2022	SHAP; LIME; XAI	Explainable forecasting	Not federated	Federated explainable analytics
20	AAAI, 2023	Adaptive model fusion	Personalized FL models	No domain explainability	Analytical explanation layer

Literature Survey (21–25)

No.	Author/Year	Method Used	Research Focus	Limitations	Research Gap
21	arXiv, 2025	Attention-LSTM; FedAvg; sliding window	Early sepsis prediction	Black-box predictions	Explainable clinical forecasting
22	MDPI, 2023	CNN-LSTM; GANs; edge FL	Multimodal sepsis detection	Heavy computation	Lightweight explainable models
23	JMIR, 2025	XGBoost; SHAP; ensemble	Explainable CDS for sepsis	Not federated	Privacy-preserving XAI + FL
24	PMC, 2025	Multi-channel FL; calibration	Hospital-scale FL systems	Complex synchronization	Automated reporting + simplification
25	IEEE Access, 2025	SMPC; encrypted FedAvg	Secure edge FL	Training delay	Explainable + LLM-based interfaces

Research Gap Analysis

Existing Research	Identified Limitations / Gap	Need / Why Gap Matters	Project Contribution
Centralized ML and on-device analytics systems (Papers 1, 17, 19, 23)	Do not support federated learning or privacy-preserving collaboration	Centralized data sharing violates privacy laws and is unsuitable for distributed environments	Enables collaborative analytics using federated learning without sharing raw data
Basic Federated Learning frameworks (FedAvg) (Papers 2, 4, 5)	Lack personalization, explainability, and analytics-level insights	A single global model performs poorly on heterogeneous client data	Introduces personalized local models with global aggregation and analytics
Personalized Federated Learning methods (FedPer, pFL) (Papers 3, 10, 16, 20)	Personalization exists but lacks explainability and drift awareness	Without explainability and adaptation, predictions cannot be trusted over time	Combines personalization with explainable analytics and drift handling
Concept drift–aware FL approaches (Papers 2, 9)	Drift handling is not integrated with analytics or forecasting pipelines	Real-world data behavior changes over time, degrading prediction accuracy	Integrates drift detection with time-series analytics and forecasting
Federated Time-Series Forecasting works (Papers 6, 8, 10–15, 18)	Primarily focus on accuracy; limited explainability and personalization	Stakeholders require transparency to trust and act on predictions	Provides explainable insights alongside federated forecasting
Explainable AI approaches (Papers 7, 17, 19, 23)	Explainability mostly exists in centralized or non-time-series settings	Explainability is critical for trust in sensitive domains like healthcare and energy	Adapts explainable AI techniques to federated time-series analytics
Healthcare-focused FL applications (Papers 9, 21, 22, 24, 25)	Application-specific solutions with limited generalization	A reusable framework is required across multiple domains	Proposes a domain-agnostic federated analytics framework
Security- and privacy-enhanced FL methods (Papers 13, 24, 25)	High complexity and low usability despite strong privacy guarantees	Practical systems must balance privacy, performance, and usability	Balances privacy, efficiency, and interpretability in a single framework

Existing Systems & Limitations

Existing System	Key Feature	Limitation
Centralized ML Systems	High accuracy with full data	Privacy risk, no data sharing
Traditional Time-Series	Simple forecasting	One model doesn't fit all
Basic Federated Learning	Privacy-preserving	No personalization, no drift

Gap Identified: No system combines privacy + personalization + explainability + drift adaptation.

Proposed Solution

Novel Federated Learning-Based System

We propose: A unified federated time-series forecasting framework that enhances FedAvg by integrating personalization, CUSUM-based drift adaptation, and attention-driven explainability for non-IID distributed environments.

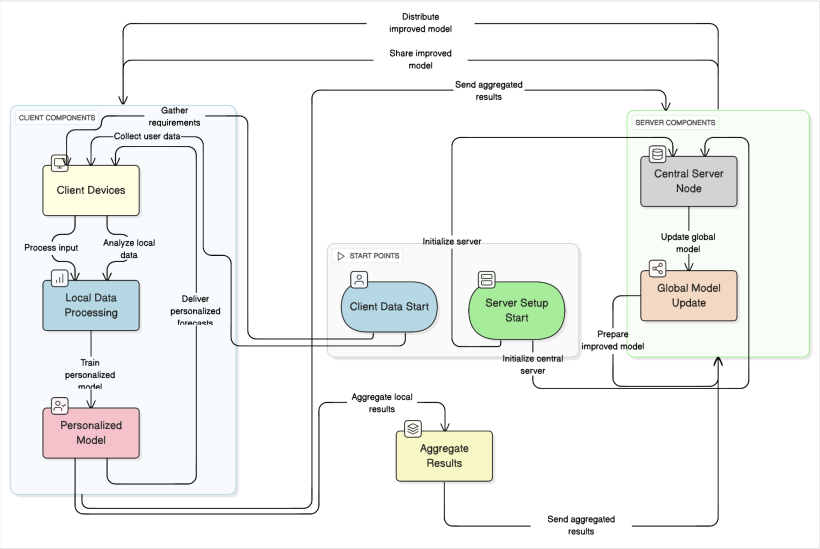
Key Idea:

Our framework jointly combines FL,personalization,drift handling,XAI to support heterogeneous clients while preserving strict data privacy. It enables interpretable, drift-aware personalized forecasting through a communication-efficient federated pipeline.

Novelty Statement

Extension : We enhance FedAvg-based federated learning by tightly integrating heterogeneity-aware personalization, CUSUM-based concept drift adaptation, and attention-driven explainability into a unified multivariate time-series forecasting framework. Unlike prior works that address these components independently, our approach delivers a privacy-preserving, adaptive, and interpretable solution

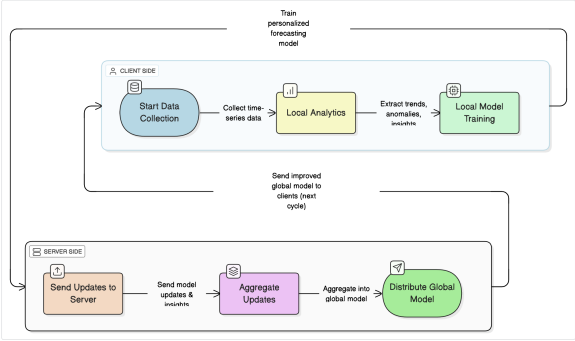
System Architecture Diagram



Workflow

End-to-End Process Flow

1. **Data Collection:** Client collects time-series data locally
2. **Local Analytics:** Extract trends, anomalies, and insights
3. **Model Training:** Personalized forecasting model trained locally
4. **Secure Sharing:** Only model updates & insights sent to server
5. **Global Aggregation:** Server aggregates updates into global model
6. **Distribution:** Improved global model sent back to all clients



Data Flow / Communication

Privacy-Preserving Architecture

Key Principles:

- Raw data stays on client devices
- Local analytics $\rightarrow$ model updates $\rightarrow$ server
- Server aggregates updates securely
- No raw data transmission
- Validation and aggregation ensure privacy

Data Flow:

Input $\rightarrow$ Local Processing $\rightarrow$ Model Update $\rightarrow$ Global Aggregation $\rightarrow$ Output

Tools, Technologies & Protocols

Technical Stack

Frontend:

- Web-based dashboard
- HTML, CSS, JavaScript

Backend:

- Python (FL logic)
- REST API

Models:

- LSTM / Transformer
- Time-series forecasting

Libraries:

- TensorFlow / PyTorch
- NumPy, Pandas

Protocols:

- REST API
- HTTP/HTTPS

Security:

- Encrypted communication
- No raw data transmission

Technical Advantages of Proposed Framework

Research-Level Contributions

Privacy-Preserving Optimization

Federated gradient aggregation ensures raw multivariate time-series data remains local while only encrypted model parameters are shared.

Attention-Based Explainability

Temporal attention mechanism computes timestep importance weights for interpretable multivariate forecasting.

Heterogeneity-Aware Personalization

Hybrid global backbone with client-specific layers handles non-IID distributions and improves localized forecasting accuracy.

Communication-Efficient Scalability

Weighted federated aggregation minimizes communication overhead while supporting multi-client deployment.

Drift-Adaptive Learning

Integrated CUSUM-based concept drift detection enables automatic retraining under non-stationary temporal dynamics.

Unified Integrated Pipeline

Single framework jointly integrates FL, personalization, drift monitoring, and XAI — eliminating isolated module design.

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Thank You

Questions?